Spring potential energy

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Learning target: Calculate stored spring energy and distinguish it from force.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 12.*
15 J, by conservation of mechanical energy.

02. Worked example: explain the model
Both triangle height and base double, so area and stored energy quadruple.

03. Read and interpret the model
Energy 4 times; force 2 times.

04. Apply the physics
$x^2=0.04\ m^2$; U 2 J.

05. Explain your reasoning
x^2 positive; force has direction, energy scalar measured relative equilibrium.

06. Review: connect an earlier idea *Review: Unit 4, Day 3.*
-24 J; the force opposes displacement.

07. Exit check
12 J within Hooke’s law range.

08. STRETCH: test the model
0.25, 1, 2.25 J; curved quadratic increase.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 12.*
No. That height requires an increase mg×2 in potential energy, exceeding the initial energy; it would require negative final K.

02. Worked example: explain the model
Both triangle height and base double, so area and stored energy quadruple.

03. Read and interpret the model
Energy 4 times; force 2 times.

04. Apply the physics
U 1 J; $v=\sqrt{(2/0.5)}=2$ m/s.

05. Explain your reasoning
x^2 positive; force has direction, energy scalar measured relative equilibrium.

06. Review: connect an earlier idea *Review: Unit 4, Day 3.*
W=-24 J because cos 120°=-1/2; the component opposes displacement.

07. Exit check
$|x|=\sqrt{(0.8/80)}=0.10$ m; force 8 N.

08. STRETCH: test the model
Energy 4.5 J; $v=\sqrt{18}\approx4.24$ m/s. Compression 0.30 m doubles speed because spring energy is quadratic.